

From BOAMP Notices to Procurement-Recurrence Risk: A Proxy-Event and Survival-Analysis Framework for Gigalis

First draft synthesis report

Generated from executed project outputs on 2026-07-08

Abstract

This synthesis turns the BOAMP analysis into one coherent argument for Gigalis. It separates the business question, the absence of official renewal-chain labels, the construction of a proxy recurrence event, the benchmark and diagnostic validation layers, the M0/M1/M2 method comparison, and the survival outputs used for 12- and 24-month operational prioritization. All reported values are traced in `source_values_used.csv`.

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1 Executive Summary

Gigalis wants to anticipate future public digital procurement needs early enough to prepare account monitoring, segment planning, and commercial responses. BOAMP is useful for this objective because it records public notices over time, but it is not a contract-history database and it does not contain an official field linking an older call for tender to a later renewal.

The current selected event definition is **M2 balanced** with match-probability threshold 0.65. The final selected survival input is the M2 balanced survival CSV listed in the trace file. It contains 1,210 eligible contracts, 254 proxy recurrence events, and 956 censored rows, giving an event rate of 21.0%. The event is an identifiable reappearance of a similar procurement need; it is not a verified legal renewal.

The selected M2 balanced method is benchmark-preferred over M0 balanced: benchmark-estimated precision/recall/F1 are 0.612/0.733/0.667, compared with 0.575/0.568/0.572 for M0 balanced. M0 balanced remains the conservative transparent baseline with 269 events and 22.2%. The old 665-event result is historical pre-calibration only.

Under M2 balanced, the Kaplan-Meier median is not reached because fewer than half of contracts experience the selected proxy event before censoring. Estimated survival is 95.6% at 12 months, 92.1% at 24 months, 84.0% at 48 months, and 76.0% at 60 months. The reduced Cox C-index is 0.553; the richer category-aware Cox specification has C-index 0.606. LogNormalAFT is the best AFT model in the method-comparison output, with AIC 3357.9. Operational indicators are available for 1,204 contracts; median p12 is 0.0206, median p24 is 0.0634, and expected proxy recurrences are 29.0 at 12 months and 84.1 at 24 months.

2 Internship and Business Context

The internship project supports Gigalis in moving from retrospective procurement observation to forward-looking monitoring. The practical question is not only whether a buyer has purchased digital services before, but when a similar need may reappear and which buyers, segments, or contracts deserve attention during the next 12 to 24 months.

The output is therefore designed as a prioritization layer. It can help analysts and account managers decide where to monitor BOAMP publications, where to prepare domain expertise, and which digital segments may require earlier commercial preparation. The result should not be read as a deterministic renewal forecast. It is a structured risk indicator for observable BOAMP proxy recurrences.

3 Data Source and Analytical Population

BOAMP notices provide the public procurement corpus. The survival unit is the **APPEL_OFFRE** notice because it represents the initial procurement opportunity whose later recurrence is being monitored. **ATTRIBUTION** notices are not treated as separate survival units; they are used for start-date refinement where available because award information can improve the timing of the original contract.

The current processed BOAMP file covers publication dates from 2015-03-02 to 2024-12-29. For the selected survival input, start dates run from 2015-03-02 to 2024-03-22. Rows are eligible when they have enough duration and timing information to define an observation window and either a selected proxy event or right-censoring.

Table 1: Analytical population funnel. Source: executed BOAMP processed and selected M2 survival outputs.

Step	Count	Share of previous	Source
Raw BOAMP notices	3,181	–	boamp_full_clean.csv
APPEL_OFFRE notices	1,933	60.8%	boamp_full_clean.csv
Eligible contracts	1,210	62.6%	M2 survival input
Selected M2 proxy events	254	21.0%	M2 survival input
Censored rows	956	79.0%	M2 survival input

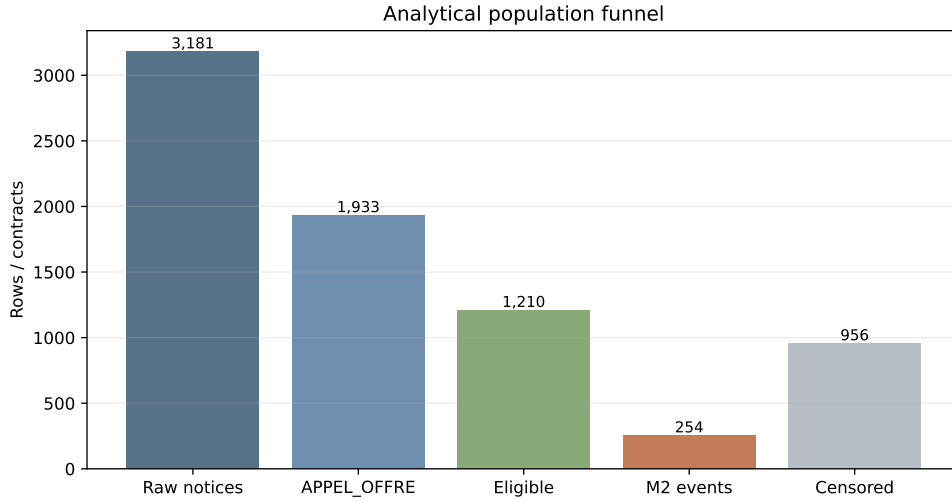


Figure 1: Dataset funnel based on real BOAMP outputs. Source: generated from the executed project pipeline and validation outputs.

4 Core Methodological Challenge: No Ground Truth

BOAMP has no official renewal-chain ground truth. There is no field that reliably says one call for tender is the renewal of another, and real BOAMP precision and recall are therefore not directly observable. This matters because a later notice from the same buyer may be a genuine reappearance of a similar need, a related but different procurement, a publication artefact, or an unrelated requirement.

In this report, **event** = 1 means a proxy recurrence under the selected rule: an identifiable reappearance of a similar procurement need. **event** = 0 means censored or no observed proxy recurrence under the available data and rule. Neither label is legal ground truth. The validation vocabulary is therefore deliberately conservative: benchmark-estimated precision/recall, diagnostic evidence, negative-control diagnostics, and method-neutral review.

5 Data Cleaning and Feature Construction

The pipeline converts notice-level BOAMP data into a contract-level survival dataset. Buyer keys are constructed from available organization identifiers and cleaned buyer names; SIREN/SIRET enrichment helps reduce buyer fragmentation but does not eliminate it. CPV codes are normalized into division and category features, and generic CPV codes are flagged because broad codes such as software or IT services can create ambiguous links.

Duration cleaning preserves the raw duration when possible, flags suspicious values, and imputes administrative durations only when needed for survival construction. Start dates are built from publication dates and refined by attribution dates where available. Sentence-Transformer embeddings provide semantic similarity between source and candidate procurement objects. The technology taxonomy is provisional and based on CPV/keyword logic; supervised NLP classification is handled separately by the team and does not block the survival pipeline.

6 Initial Baseline Method and Why It Was Replaced

The initial baseline linked notices using semantic similarity, buyer agreement, CPV evidence, and timing around the expected end date. It produced 665 events among 1,210 eligible contracts, an event rate of 55.0%. This high-linkage result was useful for exploration, but it maximized link quantity rather than analytical reliability.

Manual audit evidence showed low precision in parts of that baseline. Later analysis also showed that the manual audit sample was itself baseline-biased because it was stratified on the pre-calibration baseline’s own links and confidence tiers. The project therefore changed objective: instead of maximizing the number of links, it constructs a calibrated proxy-event definition with transparent benchmark and diagnostic evidence. The 665-event result is retained only as a historical pre-calibration baseline.

7 Synthetic BOAMP-like Benchmark

The synthetic benchmark was needed because real BOAMP labels are unavailable. It creates BOAMP-like source and candidate pairs where the matching truth is known by construction. That makes it possible to estimate precision, recall, and F1 for candidate linkage rules without pretending that real BOAMP truth is observed.

The benchmark simulates buyer-name variation, generic CPV, CPV drift, missing CPV, missing duration, paraphrased text, timing shifts, and ambiguous same-buyer candidates. It uses easy, medium, and hard scenarios. The noise audit reports buyer-name-noise rates from 11.1

Table 2: Balanced-method synthetic benchmark performance by scenario, shown as precision/recall/F1. Source: synthetic benchmark outputs.

Method	Easy	Medium	Hard	Generic CPV
M0 balanced	0.777/0.806/0.791	0.601/0.586/0.593	0.318/0.302/0.310	0.464/0.525/0.493
M1 balanced	0.812/0.924/0.865	0.619/0.764/0.684	0.409/0.501/0.450	0.487/0.615/0.544
M2 balanced	0.812/0.924/0.865	0.619/0.764/0.684	0.409/0.501/0.450	0.487/0.615/0.544

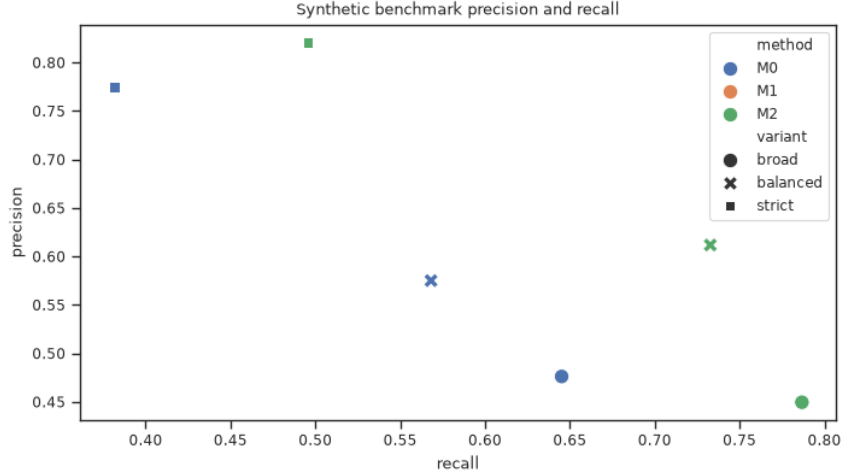


Figure 2: Synthetic precision-recall comparison for M0, M1, and M2. This is a controlled synthetic benchmark, not real BOAMP ground truth. Source: generated from the executed validation outputs.

8 Linkage Method Comparison: M0, M1, M2

M0 is the deterministic calibrated composite rule. It combines text similarity, CPV compatibility, timing, and buyer logic into transparent thresholds. M1 is a probabilistic linkage model trained on synthetic pair labels in the same feature space. M2 is an active-learning-assisted variant using the same probability feature space and producing a targeted review queue.

The current final recommendation promotes M2 balanced to the main method because all promotion criteria pass: the selection score is 0.797 versus 0.741 for M0 balanced; negative-control acceptance is lower than M0; the event count remains sufficient for survival; the synthetic-real profile shift is acceptable; and the text backend matches the real pipeline. Mapped audit precision is not used for final selection because the audit sample is baseline-biased. It remains a plausibility diagnostic only.

Table 3: Method comparison. Synthetic metrics are benchmark-estimated; real event counts are proxy recurrences in BOAMP.

Method	Prec.	Rec.	F1	Events	Event rate	Neg. control	Cox C	Role
M0 balanced	0.575	0.568	0.572	269	22.2%	9.4%	0.544	conservative baseline
M1 balanced	0.612	0.733	0.667	256	21.2%	8.5%	–	alternative
M2 balanced	0.612	0.733	0.667	254	21.0%	7.9%	0.553	selected main

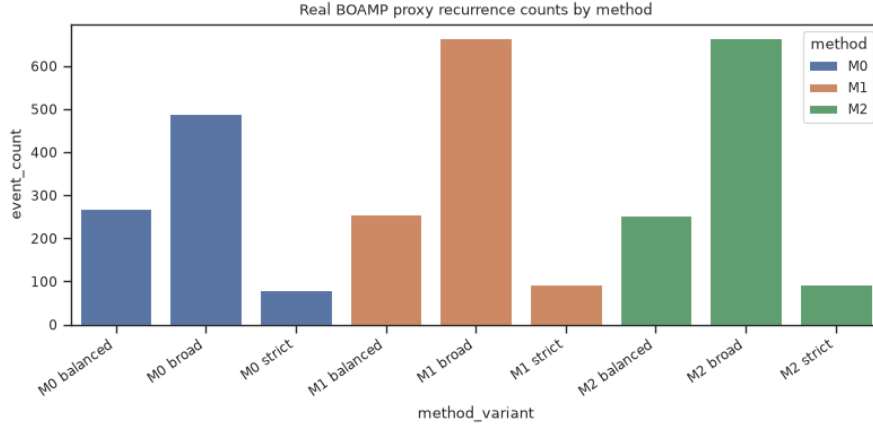


Figure 3: Real BOAMP proxy-event counts by method. Counts are constructed proxy events, not verified legal renewals. Source: generated from executed method-comparison outputs.

9 Final Selected Event Definition

The final selected definition is **M2 balanced**, a match-probability rule at threshold 0.65. The final dataset is `data/processed/boamp_phase2_survival_method_m2_balanced.csv` with 1,210 eligible contracts, 254 events, 21.0% event rate, and 956 censored rows.

M0 balanced remains the conservative transparent baseline: 269 events, 22.2%, and negative-control acceptance 9.4%. M0 strict is a high-confidence sensitivity with 79 events and 6.5%. Broad rules are high-recall sensitivity checks. The old 665-event result is historical pre-calibration only.

Real BOAMP pipeline and validation flow

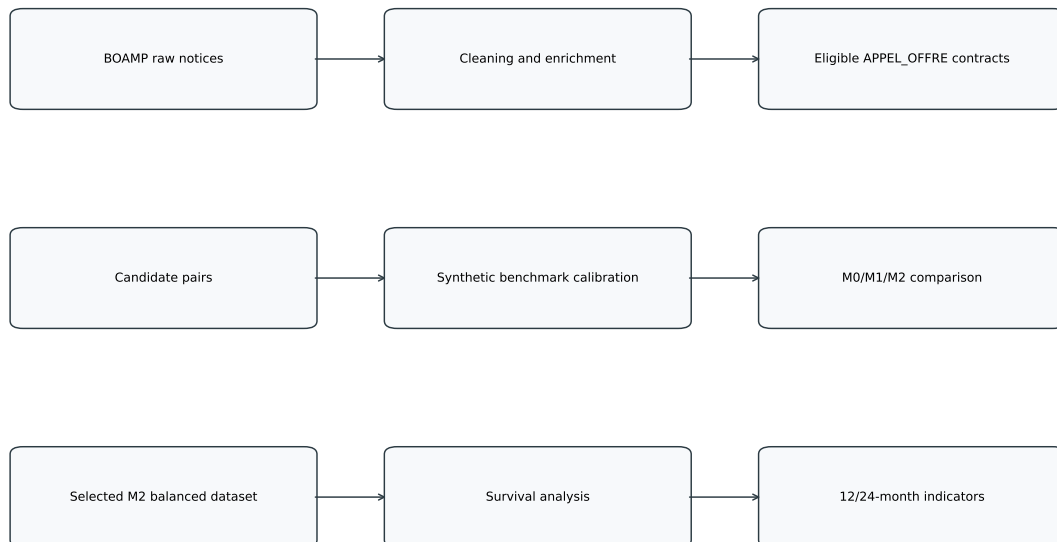


Figure 4: Real BOAMP pipeline from raw notices to operational indicators. Source: generated from the executed project pipeline and validation outputs.

10 Survival-Analysis Methodology

The selected event definition creates a right-censored survival dataset. Each row has an observed duration: either the time from source start to the selected proxy recurrence, or the time from source start to censoring when no proxy recurrence is observed. Kaplan-Meier estimates the survival curve without assuming a parametric shape. Cox PH estimates relative hazard under proportional-hazard assumptions and reports C-index as a ranking-quality diagnostic. AFT models estimate parametric survival time distributions and are compared by AIC within the same event definition.

Fixed-horizon 12- and 24-month predictions translate the survival model into operational prioritization scores. These scores estimate identifiable BOAMP proxy recurrences under the model, not legal renewal probabilities. AIC values are comparable within the same event definition and specification; they should not be used to claim that one event definition is legally truer than another.

Survival modeling workflow



Figure 5: Survival modeling workflow for the selected M2 proxy-event definition. Source: generated from the executed project pipeline and validation outputs.

11 Survival Results Under the Selected Method

Under M2 balanced, there are 254 events among 1210 contracts. The Kaplan-Meier median is not reached because fewer than 50% of contracts experience the selected proxy event before censoring. Survival is 95.6% at 12 months, 92.1% at 24 months, 84.0% at 48 months, and 76.0% at 60 months.

The reduced Cox C-index is 0.553; the richer category-aware specification reaches 0.606. PH diagnostics show a strong violation for declared duration ($p = 3.59e-25$), while imputation status ($p = 0.728$) and start year ($p = 0.406$) do not show the same violation. This supports using Cox mainly as a diagnostic ranking model and keeping AFT/fixed-horizon indicators central for operations.

The method-comparison output selects LogNormalAFT as the best AFT model with AIC 3357.9. The separate M2 AFT refit table also selects LogNormalAFT, with AIC 3360.5; this small difference is retained in `source_values_used.csv` because both values exist in executed outputs.

Table 4: Survival headline comparison. Survival probabilities come from executed method-survival outputs.

Method	Events	KM median	S12	S24	S48	S60	Cox C
M0 balanced	269	not reached	95.9%	92.3%	83.2%	75.2%	0.544
M2 balanced	254	not reached	95.6%	92.1%	84.0%	76.0%	0.553
M0 strict	79	not reached	99.0%	98.1%	95.5%	91.6%	0.538

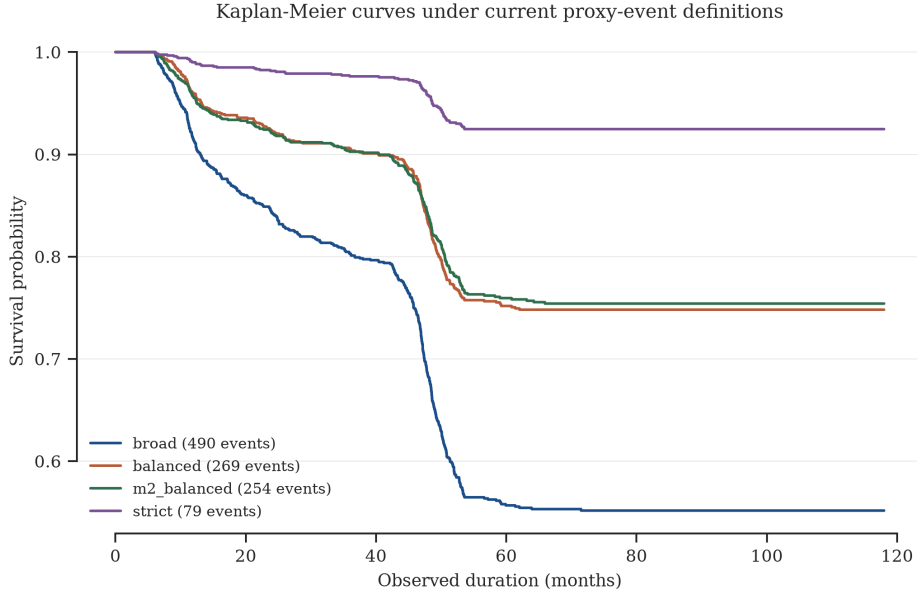


Figure 6: Kaplan-Meier comparison for real BOAMP proxy-event definitions. Source: generated from executed survival outputs.

12 Operational 12/24-Month Indicators

The selected survival model is converted into contract-level p12 and p24 scores. These are prioritization scores for identifiable BOAMP proxy recurrences. They can support monitoring lists and segment planning, especially where many small probabilities aggregate into a meaningful buyer or segment workload.

Operational indicators are available for 1,204 contracts. The median 12-month score is 0.0206; the median 24-month score is 0.0634. Summed across scored contracts, expected proxy recurrences are 29.0 at 12 months and 84.1 at 24 months.

Table 5: Top buyer-level expected proxy recurrences under M2 balanced. Source: survival model output.

Buyer key	Contracts	Expected 12m	Expected 24m
SIREN:234400034	91	2.8	7.7
NAME:nantes metropole	72	1.6	4.7
SIREN:935367433	28	1.0	2.7
SIREN:264400136	42	0.9	2.6
SIREN:391722881	28	0.9	2.4
SIREN:244900015	30	0.7	2.0

Table 6: Top segment-level expected proxy recurrences under M2 balanced. Source: survival model output.

Segment	Contracts	Expected 12m	Expected 24m
IT Services & Consulting	343	11.9	32.4
Software & Applications	223	3.7	11.6
Cybersecurity	136	3.4	9.8
Telecom & Networks	220	3.3	10.4
Unknown	114	2.7	8.0
Digital Workplace & Collaboration	51	1.2	3.5

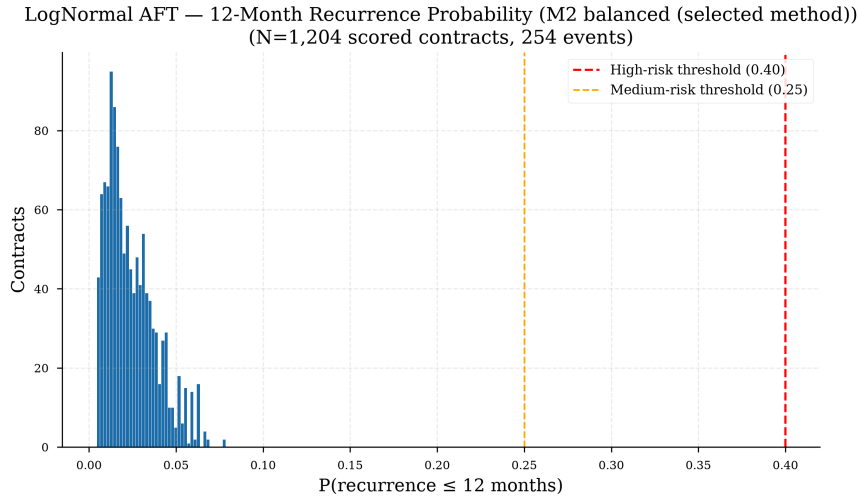


Figure 7: Distribution of 12-month proxy-recurrence prioritization scores under M2 balanced. Source: survival model output.

13 Robustness and Sensitivity

The survival conclusions are stable across M0 balanced and M2 balanced in the current outputs. M0 balanced has 95.9% 12-month survival and 92.3% 24-month survival, while M2 balanced has 95.6% and 92.1%. This supports the applied interpretation that most contracts do not show an identifiable proxy recurrence within the first two years.

M0 strict is mainly a lower-bound sensitivity: it has only 79 events and 6.5% event rate. Broad rules are useful to check high-recall behavior but are not selected for the main survival model. M2 has a higher generic-CPV share than M0 balanced (37.0% versus 23.0%), so generic-CPV M2 links should be reviewed first in the next manual validation pass.

14 Limitations

Table 7: Limitations and mitigations.

Limitation	Consequence	Mitigation / current status
No real BOAMP ground truth	Real precision and recall cannot be observed directly	Report proxy recurrence only; use method-neutral review next

Limitation	Consequence	Mitigation / current status
Synthetic benchmark is controlled, not real truth	Benchmark-estimated metrics may not transfer perfectly	Treat as calibration evidence, not proof
Manual audit baseline bias	Mapped audit precision is not a final arbiter	Use as plausibility diagnostic only
Active-learning review sample not completed	M2 review queue has not become validation labels	Complete method-neutral manual review
Generic CPV risk	Higher M2 generic-CPV share may hide weaker semantic evidence	Review generic-CPV M2 links first
Buyer fragmentation remains	Same organization can appear under multiple buyer keys	Continue SIREN/SIRET enrichment and matching checks
Declared duration is administrative and violates PH	Reduced Cox assumptions are partly strained	Prefer AFT and fixed-horizon indicators for operations
Cox C-index is weak-to-moderate	Ranking power is useful but limited	Present risk scores as prioritization, not certainty
Supervised NLP classifier handled separately	Segment labels are provisional in this survival work	Integrate final team NLP labels when available
Trend/change-point analysis remains future work	The report does not yet model demand shifts over calendar time	Add Phase 4 trend analysis later

15 Conclusion and Next Steps

The project now has a calibrated, traceable proxy-event definition for BOAMP recurrence analysis. The selected main method is M2 balanced; M0 balanced remains the conservative transparent baseline; the old 665-event output is historical pre-calibration only. Survival analysis under the selected method produces readable 12- and 24-month indicators for operational prioritization.

The next work should focus on method-neutral manual review of M2 links, inspection of generic-CPV M2 links, integration of final NLP labels when the separate team classifier is available, Phase 4 trend/change-point analysis, and optional DECP enrichment. Those steps would make the operational layer more robust without changing the central interpretation: BOAMP does not provide legal renewal ground truth, so the analysis must remain explicit about proxy recurrence and diagnostic evidence.

A Validation Framework Diagrams

No-ground-truth validation framework

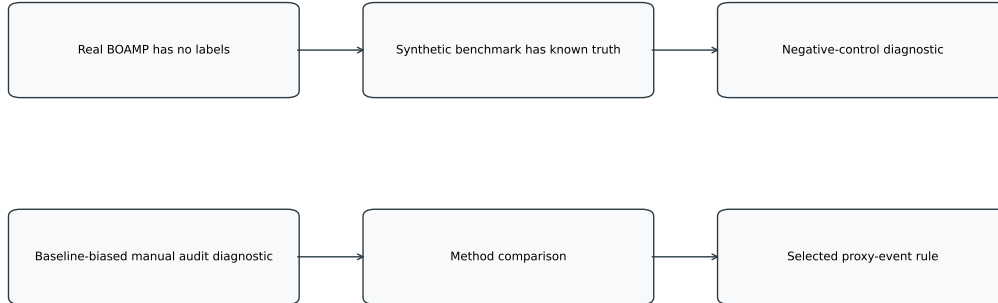


Figure 8: Validation framework in a no-ground-truth BOAMP setting. Source: generated from the executed project pipeline and validation outputs.

Proxy-event construction

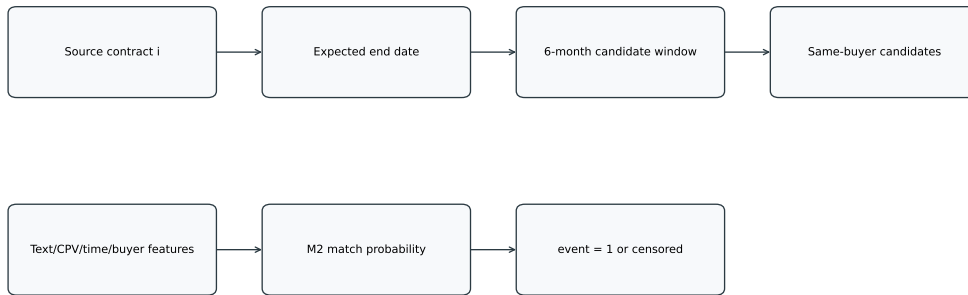


Figure 9: Proxy-event construction diagram for real BOAMP candidate links. Source: generated from the executed project pipeline and validation outputs.

Method logic

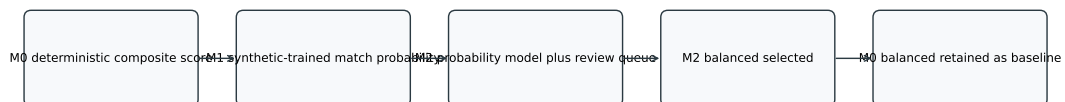


Figure 10: M0/M1/M2 method logic. Source: generated from the executed project pipeline and validation outputs.

B Top Contract Indicators

Table 8: Top contract-level p12/p24 scores under M2 balanced. Source: survival model output.

Contract	Segment	p12	p24
BOAMP:24-12133	IT Services & Consulting	0.0785	0.1839
BOAMP:24-21999	IT Services & Consulting	0.0777	0.1824
BOAMP:22-112462	IT Services & Consulting	0.0692	0.1665
BOAMP:22-153148	IT Services & Consulting	0.0677	0.1637
BOAMP:22-123056	IT Services & Consulting	0.0668	0.1621
BOAMP:23-20260	IT Services & Consulting	0.0668	0.1621
BOAMP:23-41937	IT Services & Consulting	0.0668	0.1621
BOAMP:23-92003	IT Services & Consulting	0.0668	0.1621

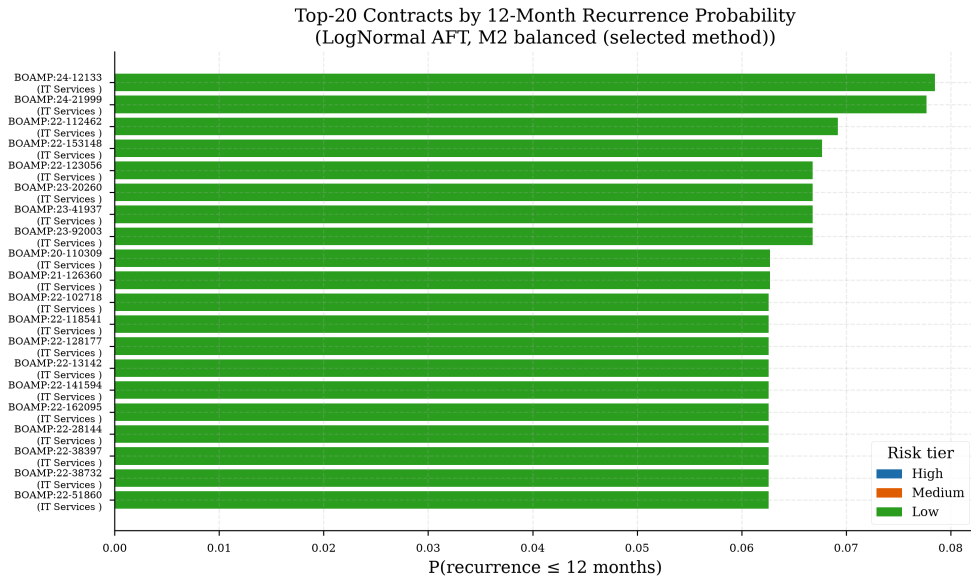


Figure 11: Top contract-level 12/24-month proxy-recurrence scores under M2 balanced. Source: survival model output.

C Source Trace Extract

The complete trace table is stored in `source_values_used.csv`. The extract below shows the main headline values and their executed output sources.

Table 9: Extract from the source trace table.

Section	Metric	Value	Source file
selection	selected method	M2 balanced	final recommendation
selection	threshold	0.65	probabilistic linkage results
population	raw BOAMP notices	3,181	cleaned BOAMP data
population	APPEL_OFFRE notices	1,933	cleaned BOAMP data
population	eligible contracts	1,210	M2 balanced survival input
population	selected M2 events	254	M2 balanced survival input
population	selected M2 censored rows	956	M2 balanced survival input
historical	pre-calibration events	665	historical handoff
method comparison	M2 synthetic precision/recall/F1	0.612 / 0.733 / 0.667	synthetic metrics
method comparison	M2 negative-control acceptance	7.9%	method comparison
survival	M2 survival 12/24/48/60m	95.6% / 92.1% / 84.0% / 76.0%	method survival comparison
survival	reduced Cox C-index	0.553	method survival comparison
survival	richer Cox C-index	0.606	category-aware Cox table
survival	best AFT / AIC	LogNormalAFT / 3357.9	method survival comparison
operational	scored contracts	1,204	M2 risk indicators
operational	median p12 / p24	0.0206 / 0.0634	M2 risk indicators
operational	expected 12/24m recurrences	29.0 / 84.1	M2 risk indicators

D Internal References

This first draft uses only existing project outputs and method names already present in the repository. The main internal sources are the final method recommendation table, the linkage method comparison tables, the method survival comparison table, the M2 balanced 12/24-month risk table, and the selected M2 balanced survival input. Exact paths are listed in `source_values_used.csv`. External literature citations can be added later if the report is expanded into a formal academic manuscript.